

CHAPTER 2 : ENERGY, ENERGY TRANSFER & GENERAL ENERGY ANALYSIS

Section	Core Revision Focus
2.1	Conservation of energy, energy quality, refrigerator example
2.2	Forms of energy, total/internal/mechanical energy, KE/PE, flow energy, mass & energy flow rates
2.3	Heat transfer, adiabatic process, specific heat transfer, heat-transfer rate, conduction/convection/radiation
2.4	Work transfer, sign convention, path functions, electrical work, generalized work
2.5	Mechanical work, shaft/spring/elastic/surface work, raising & accelerating bodies
2.6	First law, energy balance, closed/control-volume systems, stationary systems, rate/specific forms, cycles
2.7	Efficiency, HHV/LHV, combustion/power-plant/mechanical/pump/turbine/motor/generator efficiencies, combined efficiencies
2.8	Air pollution, smog/ozone, CO, particulate matter, acid rain, greenhouse effect, climate change, CO ₂ emissions, EVs, renewable energy

2.1 INTRODUCTION

- **Energy conservation:** Total energy is conserved during a process.
- **Energy quality:** High-quality energy (e.g. electrical energy) can be completely converted to low-quality thermal energy, but the reverse conversion is limited.
- Refrigerator in an insulated room: electrical energy supplied ultimately increases the internal energy and temperature of the room.

2.2 FORMS OF ENERGY

TOTAL ENERGY

$$e = \frac{E}{m}$$

- E : Total energy
- e : Specific total energy
- m : Mass

Only changes in energy are required; the reference level for absolute energy may be chosen arbitrarily.

Energy classification:

- **Macroscopic energy:** Associated with the system as a whole relative to an external reference frame.
- **Microscopic energy:** Associated with molecular structure and molecular activity.
- Sum of microscopic energies = **internal energy** U .

KINETIC ENERGY

$$KE = \frac{mV^2}{2} \quad ke = \frac{V^2}{2}$$

For a rotating solid body:

$$KE = \frac{1}{2}I\omega^2$$

POTENTIAL ENERGY

$$PE = mgz \quad pe = gz$$

z is measured relative to a selected reference level.

TOTAL ENERGY OF A SIMPLE COMPRESSIBLE SYSTEM

When magnetic, electric and surface-tension effects are negligible:

$$E = U + KE + PE$$

$$E = U + \frac{mV^2}{2} + mgz$$

On a unit-mass basis:

$$e = u + \frac{V^2}{2} + gz$$

For a stationary system:

$$\Delta KE = 0, \quad \Delta PE = 0, \quad \boxed{\Delta E = \Delta U}$$

INTERNAL ENERGY

Internal energy includes microscopic kinetic and potential energies.

- **Sensible energy:** Associated with molecular kinetic energy; generally increases with temperature.
- **Latent energy:** Associated with phase of a substance.
- **Chemical energy:** Associated with atomic bonds.
- **Nuclear energy:** Associated with strong binding forces within the nucleus.

MECHANICAL ENERGY

Mechanical energy can be completely and directly converted into mechanical work by an ideal mechanical device.

$$\boxed{\text{Mechanical energy} = \text{flow} + \text{kinetic} + \text{potential}}$$

For a flowing fluid:

$$e_{\text{flow}} = \frac{P}{\rho}$$

$$e_{\text{mech}} = \frac{P}{\rho} + \frac{V^2}{2} + gz$$

$$\dot{E}_{\text{mech}} = \dot{m} \left(\frac{P}{\rho} + \frac{V^2}{2} + gz \right)$$

For an incompressible fluid:

$$\Delta e_{\text{mech}} = \frac{P_2 - P_1}{\rho} + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)$$

$$\Delta \dot{E}_{\text{mech}} = \dot{m} \left[\frac{P_2 - P_1}{\rho} + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right]$$

- Constant $P, \rho, V, z \Rightarrow \Delta e_{\text{mech}} = 0$.
- Positive $\Delta e_{\text{mech}} \Rightarrow$ mechanical work supplied to fluid.
- Negative $\Delta e_{\text{mech}} \Rightarrow$ mechanical work extracted from fluid.

For an ideal turbine:

$$\dot{W}_{\max} = \dot{m} \Delta e_{\text{mech}}$$

Mass flow rate:

$$\dot{m} = \rho \dot{V}$$

$$\dot{V} = A_c V_{\text{avg}}$$

$$\dot{m} = \rho A_c V_{\text{avg}}$$

Energy flow rate:

$$\dot{E} = \dot{m} e$$

$$1 \text{ kJ/s} = 1 \text{ kW}$$

2.3 ENERGY TRANSFER BY HEAT

HEAT TRANSFER

Heat = energy transferred solely because of a temperature difference.

- Heat flows from higher temperature to lower temperature.
- No temperature difference $\Rightarrow$ no heat transfer.
- Heat is **energy in transition**; a system does not contain heat as a property.
- Heat transfer stops at thermal equilibrium.

ADIABATIC PROCESS

$$\boxed{Q = 0}$$

Occurs when:

- the system is well insulated, or
- system and surroundings are at the same temperature.

Important: Adiabatic $\neq$ isothermal. Temperature may change during an adiabatic process because of work or other energy interactions.

AMOUNT OF HEAT TRANSFER

$$q = \frac{Q}{m}$$

$$\dot{Q} = \frac{dQ}{dt}$$

For variable heat-transfer rate:

$$Q = \int_{t_1}^{t_2} \dot{Q} dt$$

For constant rate:

$$Q = \dot{Q} \Delta t, \quad \Delta t = t_2 - t_1$$

$$q = \frac{Q}{m}, \quad \dot{Q} = \frac{Q}{\Delta t}$$

MECHANISMS OF HEAT TRANSFER

- **Conduction:** Molecular interaction between adjacent particles.
- **Convection:** Energy transfer between a solid surface and moving fluid; includes conduction and fluid motion.

- **Radiation:** Electromagnetic-wave/photon transfer.

All require a temperature difference and proceed from higher to lower temperature.

2.4 ENERGY TRANSFER BY WORK

WORK

Work is energy transfer associated with a force acting through a distance.

$$\omega = \frac{W}{m}$$

$$\dot{W} = \frac{dW}{dt}$$

Sign convention:

- Heat **to** system: positive.
- Heat **from** system: negative.
- Work **by** system: positive.
- Work **on** system: negative.

Alternatively, use positive in/out quantities explicitly:

$$W_{\text{in}} > 0, \quad W_{\text{out}} > 0$$

A negative calculated value indicates that the actual direction is opposite to the assumed direction.

HEAT AND WORK AS PATH FUNCTIONS

- Heat and work exist only while crossing the system boundary.
- A system possesses energy, but not heat or work.
- Heat/work are associated with a **process**, not a state.
- Their magnitude depends on the path.
- Heat and work = **path functions**.
- Thermodynamic properties = **point functions**.

For a property:

$$\int_1^2 dV = V_2 - V_1 = \Delta V$$

For work:

$$\int_1^2 \delta W = W_{12} \neq W_2 - W_1$$

d = exact differential of a property; δ = inexact differential of a path function.

ELECTRICAL WORK

$$W_e = VN$$

$$\dot{W}_e = VI$$

For variable V and I :

$$W_e = \int_1^2 VI \, dt$$

For constant V, I :

$$W_e = VI\Delta t$$

$$\dot{W}_e = VI = I^2 R = \frac{V^2}{R}$$

GENERALIZED WORK

$$\delta W = F \, dx$$

General concept:

Generalized force $\times$ generalized displacement

- Mechanical: force $\times$ displacement.
- Electrical: voltage $\times$ charge.
- Magnetic: magnetic field strength $\times$ total magnetic dipole moment.
- Electrical polarization: electric field strength $\times$ polarization.

2.5 MECHANICAL FORMS OF WORK

Mechanical work:

$$W = Fs$$

For variable force:

$$W = \int_1^2 F ds$$

Requirements for mechanical work:

1. A force must act on the system boundary.
2. The system boundary must move.

No boundary displacement $\Rightarrow$ no mechanical work. Expansion into a vacuum $\Rightarrow$ no work because no opposing/driving force exists.

Type	Important relations	Key conditions / quantities
Shaft Work	$T = Fr$ $s = 2\pi rn$ $W_{sh} = 2\pi nT$ $\dot{W}_{sh} = 2\pi \dot{n}T$	T = torque, F = tangential force, r = moment arm, n = revolutions, $\dot{n}$ = revolutions/time. Used for engines, turbines, pumps, compressors.
Spring Work	$\delta W_{spring} = F dx$ $F = kx$ $\delta W_{spring} = kx dx$ $W_{spring} = \frac{1}{2}k(x_2^2 - x_1^2)$	Linear elastic spring; k = spring constant. x_1, x_2 measured from the undisturbed position.
Elastic Solid Bar	$W_{elastic} = \int_1^2 F dx$ $\sigma_n = \frac{F}{A}$ $W_{elastic} = \int_1^2 \sigma_n A dx$	Within elastic range, a solid bar behaves similarly to a linear spring. σ_n = normal stress.
Liquid Film	$W_{surface} = \int_1^2 \sigma_s dA$ $dA = 2b dx$ $F = 2b\sigma_s$	σ_s = surface tension, b = movable-wire length. Factor 2 because a liquid film has two surfaces.
Raising a Body	$W = \Delta PE$ $W = mg(z_2 - z_1)$	Work transferred to the body equals its increase in gravitational potential energy.
Accelerating a Body	$W = \Delta KE$ $W = \frac{1}{2}m(V_2^2 - V_1^2)$	Work required for acceleration equals change in kinetic energy.

2.6 THE FIRST LAW OF THERMODYNAMICS

First law:

$$\text{Energy in} - \text{Energy out} = \text{Change in stored energy}$$

Special cases:

$$Q = 0 \Rightarrow \Delta E = -W_{\text{net, out}}$$

$$W = 0 \Rightarrow \Delta E = Q_{\text{net, in}}$$

GENERAL ENERGY BALANCE

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

$$\Delta E_{\text{system}} = E_2 - E_1$$

For a simple compressible system:

$$\Delta E = \Delta U + \Delta KE + \Delta PE$$

$$\Delta U = m(u_2 - u_1)$$

$$\Delta KE = \frac{1}{2}m(V_2^2 - V_1^2)$$

$$\Delta PE = mg(z_2 - z_1)$$

Therefore:

$$\Delta E = m(u_2 - u_1) + \frac{1}{2}m(V_2^2 - V_1^2) + mg(z_2 - z_1)$$

STATIONARY SYSTEMS

$$\Delta KE = 0, \quad \Delta PE = 0$$

Hence:

$$\Delta E = \Delta U$$

ENERGY-TRANSFER MECHANISMS

Across a system boundary, energy can be transferred by:

1. Heat
2. Work
3. Mass flow

For a **closed system**: heat + work only.

For a **control volume**: heat + work + mass flow.

Net heat:

$$Q_{\text{net},\text{in}} = Q_{\text{in}} - Q_{\text{out}}$$

Net work:

$$W_{\text{net},\text{out}} = W_{\text{out}} - W_{\text{in}}$$

Complete energy balance:

$$\Delta E_{\text{system}} = E_{\text{in}} - E_{\text{out}} = (Q_{\text{net},\text{in}}) - (W_{\text{net},\text{out}}) + (E_{\text{mass},\text{in}} - E_{\text{mass},\text{out}})$$

$$\Delta E_{\text{system}} = E_{\text{in}} - E_{\text{out}} = (Q_{\text{in}} - Q_{\text{out}}) - (W_{\text{out}} - W_{\text{in}}) + (E_{\text{mass},\text{in}} - E_{\text{mass},\text{out}})$$

Useful eliminations:

$$\text{Adiabatic: } Q = 0 \quad \text{No work: } W = 0 \quad \text{Closed system: } E_{\text{mass}} = 0$$

RATE AND SPECIFIC FORMS

Rate form:

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \frac{dE_{\text{system}}}{dt}$$

For constant rates:

$$Q = \dot{Q}\Delta t, \quad W = \dot{W}\Delta t, \quad \Delta E = \frac{dE}{dt}\Delta t$$

Specific basis:

$$e = \frac{E}{m}$$

$$e_{\text{in}} - e_{\text{out}} = \Delta e_{\text{system}}$$

Differential form:

$$\delta e_{\text{in}} - \delta e_{\text{out}} = de_{\text{system}}$$

ENERGY BALANCE FOR A CYCLE

For a cycle:

$$\Delta E_{\text{system}} = 0$$

Thus:

$$E_{\text{in}} = E_{\text{out}}$$

For a closed-system cycle:

$$W_{\text{net},\text{out}} = Q_{\text{net},\text{in}}$$

and in rate form:

$$\dot{W}_{\text{net},\text{out}} = \dot{Q}_{\text{net},\text{in}}$$

2.7 ENERGY CONVERSION EFFICIENCIES

General definition:

$$\eta = \frac{\text{Desired output}}{\text{Required input}}$$

$$\eta(\%) = 100\eta$$

Always define clearly what constitutes the desired output and required input.

COMBUSTION EQUIPMENT

$$\eta_{\text{comb,equip}} = \frac{Q_{\text{useful}}}{\text{HV } \dot{m}_{\text{fuel}}}$$

Applies to furnaces, boilers and heaters.

Heating values:

- **LHV:** Water leaves as vapor; latent heat not recovered.
- **HHV:** Water is condensed; latent heat recovered.

$$\text{HHV} > \text{LHV}$$

For gasoline:

$$\text{LHV} \approx 44,000 \text{ kJ/kg}$$

$$\text{HHV} \approx 47,300 \text{ kJ/kg}$$

AFUE accounts for combustion efficiency plus heat losses, start-up losses and cool-down losses.

POWER-PLANT EFFICIENCIES

Thermal efficiency:

$$\eta_{\text{thermal}} = \frac{\dot{W}_{\text{net,out}}}{\dot{Q}_{\text{in}}}$$

Generator efficiency:

$$\eta_{\text{generator}} = \frac{\dot{W}_{\text{elect,out}}}{\dot{W}_{\text{shaft,in}}}$$

Overall:

$$\eta_{\text{overall}} = \eta_{\text{comb,equip}} \eta_{\text{thermal}} \eta_{\text{generator}}$$

or

$$\eta_{\text{overall}} = \frac{\dot{W}_{\text{net,electric}}}{\text{HHV } \dot{m}_{\text{fuel}}}$$

ELECTRICITY-TO-LIGHT CONVERSION

$$\eta_{\text{light}} = \frac{\text{Energy converted to light}}{\text{Electrical energy consumed}}$$

Lighting efficacy:

$$\text{Lighting efficacy} = \frac{\text{Light output (lumens)}}{\text{Electrical power input (W)}}$$

Unit: lm/W.

COOKING EFFICIENCY

$$\eta_{\text{cooking}} = \frac{\text{Useful energy transferred to food}}{\text{Energy consumed by appliance}}$$

Convection and microwave ovens are generally more efficient than conventional ovens.

MECHANICAL EFFICIENCY

$$\eta_{\text{mech}} = \frac{E_{\text{mech,out}}}{E_{\text{mech,in}}}$$

$$\eta_{\text{mech}} = 1 - \frac{E_{\text{mech,loss}}}{E_{\text{mech,in}}}$$

Losses primarily arise from friction and other irreversibilities.

PUMP EFFICIENCY

$$\eta_{\text{pump}} = \frac{\Delta \dot{E}_{\text{mech,fluid}}}{\dot{W}_{\text{shaft,in}}}$$

$$\eta_{\text{pump}} = \frac{\dot{W}_{\text{pump,u}}}{\dot{W}_{\text{pump}}}$$

TURBINE EFFICIENCY

$$\eta_{\text{turbine}} = \frac{\dot{W}_{\text{shaft,out}}}{|\Delta \dot{E}_{\text{mech,fluid}}|}$$

Absolute value is required because fluid mechanical energy decreases through a turbine.

MOTOR AND GENERATOR EFFICIENCY

Motor:

$$\eta_{\text{motor}} = \frac{\dot{W}_{\text{shaft,out}}}{\dot{W}_{\text{elect,in}}}$$

Generator:

$$\eta_{\text{generator}} = \frac{\dot{W}_{\text{elect,out}}}{\dot{W}_{\text{shaft,in}}}$$

Motor: Electrical $\rightarrow$ Mechanical

Generator: Mechanical $\rightarrow$ Electrical

COMBINED EFFICIENCIES

Pump-motor:

$$\eta_{\text{pump-motor}} = \eta_{\text{pump}}\eta_{\text{motor}}$$

$$\eta_{\text{pump-motor}} = \frac{\dot{W}_{\text{pump,u}}}{\dot{W}_{\text{elect,in}}} = \frac{\Delta\dot{E}_{\text{mech,fluid}}}{\dot{W}_{\text{elect,in}}}$$

Turbine-generator:

$$\eta_{\text{turbine-gen}} = \eta_{\text{turbine}}\eta_{\text{generator}}$$

$$\eta_{\text{turbine-gen}} = \frac{\dot{W}_{\text{elect,out}}}{\dot{W}_{\text{turbine,e}}} = \frac{\dot{W}_{\text{elect,out}}}{|\Delta\dot{E}_{\text{mech,fluid}}|}$$

2.8 ENERGY AND ENVIRONMENT

Major environmental effects of fossil-fuel combustion:

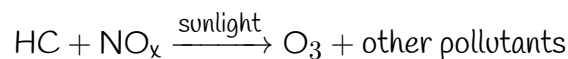
- Air pollution
- Smog
- Acid rain
- Global warming
- Climate change

Major motor-vehicle pollutants:

- Hydrocarbons (HC)
- Nitrogen oxides (NO_x)
- Carbon monoxide (CO)
- Particulate matter (PM)

OZONE AND SMOG

Ground-level ozone is the primary component of urban smog.



Hot, calm conditions favor smog formation; winds can transport pollutants over large distances.

CARBON MONOXIDE

CO is a colorless, odorless, poisonous gas produced mainly by motor vehicles.

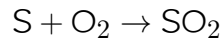
- Reduces blood's oxygen-carrying ability.
- Low concentrations can impair reactions, reflexes and judgment.
- High concentrations can be fatal.
- Particularly dangerous in enclosed/poorly ventilated spaces.

PARTICULATE MATTER

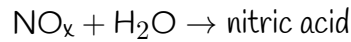
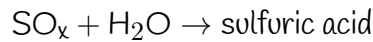
Suspended dust and soot can be emitted by vehicles, industrial facilities and combustion systems. Effects include irritation of eyes/lungs and transport of harmful substances such as acids and metals.

ACID RAIN

Sulfur combustion:



Atmospheric reactions:



These acids reaching the earth through rain/snow produce **acid rain**.

Reduction methods:

- Sulfur-dioxide scrubbers
- Low-sulfur coal
- Coal desulfurization

GREENHOUSE EFFECT AND CLIMATE CHANGE

Greenhouse effect: Certain gases absorb outgoing infrared radiation and re-emit part of it toward earth.

Major greenhouse gases:

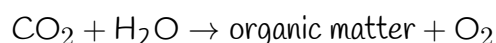
- CO_2
- Water vapor
- Methane
- Nitrogen oxides

The natural greenhouse effect is necessary for life; excessive greenhouse-gas concentrations increase trapped energy and contribute to global warming.

Major sources:

- Power generation
- Transportation
- Buildings
- Manufacturing
- Fossil-fuel combustion

Photosynthesis:



Deforestation decreases carbon uptake while fossil-fuel use increases CO_2 emissions.

Consequences:

- Weather-pattern changes
- Severe storms, heavy rainfall and flooding
- Drought
- Polar-ice melting and sea-level rise
- Ecosystem disruption
- Changes in water availability and disease distribution
- Human-health and socioeconomic impacts

International efforts:

- 1992 United Nations climate agreement
- 1997 Kyoto Protocol
- 2015 Paris Agreement

Paris Agreement: limit warming to less than 2°C above preindustrial levels while pursuing further limitation.

CARBON DIOXIDE EMISSIONS

Fossil-fueled power plant:

0.6–1.0 kg CO₂/kWh

Gasoline:

≈ 2.5 kg CO₂/L

≈ 20 lbm CO₂/gal

For a vehicle:

$$\text{CO}_2 \text{ emission} = \text{Fuel consumed} \times \text{CO}_2 \text{ emission per unit fuel}$$

Therefore:

$$\text{Higher efficiency} \rightarrow \text{Lower fuel consumption} \rightarrow \text{Lower emissions}$$

ELECTRIC VEHICLES

- No direct tailpipe emissions during operation.
- Not necessarily zero total emissions because electricity generation may involve fossil fuels.
- Environmental benefit depends on the electricity source.

Electric vehicle $\neq$ necessarily zero total emissions

Emission-free electricity sources listed in the notes:

- Hydroelectric
- Solar
- Wind
- Geothermal

RENEWABLE ENERGY AND THERMODYNAMICS

Important renewable sources:

- Hydroelectric
- Solar
- Wind
- Geothermal

Thermodynamic improvement:

Improved efficiency $\rightarrow$ Less fuel consumption $\rightarrow$ Reduced pollution

Overall exam chain:

Energy conversion $\rightarrow$ Efficiency $\rightarrow$ Fuel consumption $\rightarrow$ Environmental impact
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